COVID-19 GLOBAL IMPACT DASHBOARD

Day 4 Project Report | Interactive Dashboard Design and Analysis

1. Objective

The objective of Day 4 is to develop an interactive Power BI dashboard that helps users explore reported COVID-19 cases, deaths, country-level rates, and vaccination coverage across dates. The dashboard is designed to make time trends and country comparisons easier to understand.

2. Dataset and Tools

Dataset	Our World in Data COVID-19 data, using the Day 3 analysis CSV.
Dataset dimensions	395,311 rows and 72 columns before additional Day 3 calculations (as observed in the notebook).
Tools	Power BI Desktop, CSV, Python, Pandas, Matplotlib/Seaborn, Git and GitHub.
Key fields	location, date, total_cases, total_deaths, cases_per_million, deaths_per_million, vaccinated_pct, fully_vaccinated_pct, and 7-day averages.

3. Planned Dashboard Pages and Visuals

- **COVID-19 Overview:** KPI cards, reported cases trend, reported deaths trend, and country comparison.
- **Vaccination Analysis:** vaccination coverage trend, fully vaccinated trend, country comparison, location slicer, and date-range slicer.
- **Interactivity:** country and date filters allow users to explore selected places and time periods.

4. Preliminary Findings from the Day 3 Summary

- In the displayed country comparison, the United States had about 305,763.91 reported cumulative cases per million and 3,527.05 reported deaths per million.
- Brazil's displayed values were about 174,220.01 cases per million and 3,260.90 deaths per million.
- India's displayed vaccination record showed approximately 72.50% with at least one dose and 67.18% fully vaccinated.

5. Limitations and Interpretation

These are values from the displayed Day 3 summary, not a complete synchronized comparison for every country. The records have different latest dates and some vaccination or epidemiological fields are missing. For fair comparison, use a common date or a clearly defined latest-available record per country. Reported cases and deaths do not represent every infection or death, and a trend comparison alone does not prove causation.

6. Day 4 Deliverables Checklist

Item	Status / next action
Power BI overview page	Create and validate KPI cards and trend visuals
Vaccination analysis page	Create two line charts, country bar chart, and slicers
Validation	Check date types, missing values, and filter interactions
Documentation	Save screenshots and update README.md
GitHub	Commit and push the PBIX/report files; exclude large raw CSV files

Note: This report documents the Day 4 dashboard plan and preliminary values shared during analysis. Mark a deliverable as completed only after the visual has been built and verified in Power BI.